INDIVIDUAL WEEKLY PROJECT REPORTS

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Role: Documentation and Presentation
Project Title: Flight Delay Analysis and Visualization

WEEK 3 – Documentation of Data Preprocessing and Data Quality

During the third week, I focused on documenting the data preprocessing and data quality assessment activities carried out by the team. Since raw datasets may contain inconsistencies, missing values, duplicate records, and invalid entries, preprocessing was an important stage before performing detailed analysis.

I coordinated with the members responsible for data collection and preprocessing to understand the issues identified in the dataset. I recorded the different data quality checks performed, including checking for missing values, duplicate records, incorrect data types, and unusual numerical values.

One of the important observations discussed during this stage was the presence of negative values in the CRS elapsed time field. Such values required further examination because they could indicate invalid records or inconsistencies in the dataset. I included this observation in the documentation as an example of why data validation was necessary.

I also worked on structuring the Data Cleaning section of the project report. Rather than simply listing the cleaning activities, I focused on explaining why each activity was necessary and how it contributed to improving the reliability of the dataset.

Another part of my work involved maintaining consistency between the information collected from different team members. I organized the preprocessing details into a logical sequence so that the report could clearly explain how the raw data was examined and prepared for further analysis.

Outcome and Learning:

By the end of this week, I had documented the major preprocessing activities and understood how data quality affects the accuracy of analytical results. I also learned the importance of presenting technical processes in a clear and understandable manner.